MILITARY SPECIFICATION

SUTURE, NONABSORBABLE, SURGICAL, SILK, BRAIDED, WITHOUT NEEDLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers non-sterile, nonabsorbable, braided silk surgical sutures, without needles attached.

1.2 Classification. The sutures shall be of the following sizes:

USP	Metric
0	3.5
2	5
3	6

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-plywood.
PPP-B-621	Box, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-up.

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.
The Pharmacopeia of the United States of America (USP)

(Applications for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

3. REQUIREMENTS

3.1 Material. The nonabsorbable surgical suture shall be made of silk and as described in the current edition of the Pharmacopeia of the United States (USP). Silk shall consist of the thread prepared from the degummed cocoon filaments of glutinous gum which are secreted or spun by the mulberry silkworm, Bombyx mori (Fam. Bombycidae).

3.2 Description. The suture shall be non-sterile, in a 100-yard length evenly wound on a spool in such a manner as to permit its ready dispensing. The spool shall be of suitable construction and composition, and of sufficient size for the purpose intended.

3.2.1 Composition. The suture shall be braided and uncoated black silk. All silk sutures shall be treated to be noncapillary, and when tested as specified in 4.4.1, there shall be no evidence of capillary action. Silk sutures shall be braided silk in accordance with requirements for braided silk (USP "Class I") contained in the USP.

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3.2.2 Suture size diameter. The suture size diameter shall be determined as specified in 4.4.2 and in accordance with the USP, with the exception of the paragraph entitled Diameter, and that portion of the table in the USP referring to Limits on Diameter, which shall be substituted by Table I for each respective size.

Table I. Diameter (mm) - Range

Size	Lower Control Point	Mid-Point	Upper Control Point
1-0	0.35	0.375	0.399
2	0.50	0.550	0.599
3	0.60	0.650	0.699

3.2.3 Tensile strength. Tensile strength shall be determined as specified in 4.4.3. For Class I silk sutures delete that portion of the table in the USP paragraph entitled "Tensile Strength" listing limits on knot-pull tensile strength and substitute the following for each respective size:

Table II. Limits on Knot-Pull Tensile Strength (Kg) - Class I

Size	(A) - Average of 10 strands	(B) - Minimum per strand
1-0	2.70	1.75
2	4.40	2.50
3	5.00	3.00

Failure of sutures to comply with requirements of either average of 10 strands or minimum per strand, or both, shall be cause for rejection of the lot being tested.

3.2.4 Suture length. The length shall be measured in accordance with the requirements of the USP as specified in 4.4.4. Each suture strand shall be one hundred yards in length.

3.3 Labels. There shall be a commercial label on each spool identifying the suture as to its composition, color, capillarity, size, length, name and address of manufacturer, the legend "braided", item identification, National Stock Number (NSN) and other requirements of statute. As an alternate, the registered trademark may be substituted for name of manufacturer. The label shall be visible through the outer container. As an alternate, the National Stock No. may be present on the unit package by means of a pressure sensitive adhesive label.

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3.4 Requirements of statutes. Compliance with all requirements, statutes and regulations, promulgated thereunder, is mandatory for the contents, immediate container, labeling, and marking of any commodity governed by the requirements of any of the following: U.S. Pharmacopeia, National Formulary (N.F.), National Institutes of Health, Food and Drug Administration, Agricultural Research Service, Internal Revenue Service, Interstate Commerce Commission, or any other Federal or state laws or regulations. The sutures shall conform to the standards and or requirements of the applicable regulations or laws. Any inconsistency which exists or may appear to exist between the requirements of the foregoing and the requirements of the procurement document shall be brought to the attention of the Defense Personnel Support Center for resolution.

3.4.1 Ingredients. The ingredients entering into the preparation or manufacture of a product shall comply with the tests, standards, and requirements of the USP or N.F., and the procurement document. If the ingredients are not monographed in the USP or N.F., and standards for the ingredients are not included in the procurement document, the ingredients shall be of a high quality and purity and suitable for use in the product.

3.4.2 Testing of ingredients. Ingredients that are official in the USP or N.F. shall be tested by the methods described in the USP or N.F. and the procurement document, and the ingredients shall comply with 3.5.1. Ingredients that are not official in the USP or N.F. shall be tested by the methods specified in the procurement document or, if not specified therein, the nonofficial ingredients shall be tested by suitable methods to determine their identity and compliance with the above.

3.4.3 USP and N.F. monographed ingredients and end items. Unless otherwise specified in the schedule or in the applicable specifications, end items deliverable hereunder, as well as ingredients of such end items, shall be subject to assay and test procedures described in the applicable monographs of the USP and N.F. Alternative methods not described in the USP and N.F. are not permitted.

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3.4.4 Commingling of ingredients. The Government must be able to isolate specific lots of the active ingredients and the production of such active ingredients contained in material being purchased, to facilitate the investigation of complaints or recalls received after delivery and acceptance, and to minimize the quantities suspended pending completion of testing. Therefore, the commingling of the same active ingredient produced by different manufacturing facilities in any one production lot is prohibited.

3.5 Workmanship. Workmanship shall be first class throughout. Sutures shall be smooth, free from kinks, splinters, fraying, unraveling, cuts or knots. The entire assembly, including sutures, spools and container, shall be free from defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the silk, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated

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to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table III. The unit of product for examination shall be one spool.

Table III. Sampling for Examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table IV.

Table IV. Sampling for Tests

Characteristic	Requirement	Test Procedure	Unit of Product	Inspection level	AQL
Noncapillary	3.2.1	4.4.1	1-spool	S-2	0
Diameter	3.2.2	4.4.2	1-spool	See 4.4.2	0
Tensile Strength	3.2.3	4.4.3	1-spool	See 4.4.3	0
Length	3.2.4	4.4.4	1-spool	S-2	0

4.2.3 Sample unit usage. Where possible, the same sample unit shall be used for the determination of the two or more test characteristics.

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4.3 Examination.

4.3.1 Examination of sutures. The sutures shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables III and V. Examination, however, shall not be restricted to defects listed in Table V. Any sutures in the sample containing one or more defects shall be rejected, and if the number of defective sutures in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table V. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Required label for spool incorrect, not legible, incomplete, or missing.
102	Suture not flexible.
103	Surface of suture not smooth and free of fraying, unraveling, kinks, cuts or knots.
104	Suture strand not one continuous piece.
105	Improperly or not firmly wound, resulting in kinks, entangling or slippage during unwinding.
106	Not free from foreign matter.
Minor	
201	Color other than specified.

4.3.2 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table VI. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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Table VI. Examination of Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Except as hereinafter specified, all tests shall be conducted as specified in the USP.

4.4.1 Capillarity test. At least two 3- or 4-inch lengths of the suture shall be boiled in distilled water in a glass container for three successive periods of 20 minutes each. After the third boiling, they shall be removed, laid flat, and conditioned for 8 hours + 10 minutes in an atmosphere having a relative humidity of 65 + 2 percent at a temperature of $21^{\circ} \pm 1.1^{\circ}$ C. ($70^{\circ} \pm 2^{\circ}$ F.). After conditioning, a piece of white silk thread shall be knotted on one end of the suture and the ends of the knot cut close. The suture shall then be suspended in a dye solution of 0.5 percent methylene blue in such a manner that the white silk knot on the strand under test projects 3/8 inch above the top of the liquid. After a period of 24 hours any evidence of the dye being carried up the strand and through the white silk knot, where it is readily visible, will be considered capillary action.

4.4.2 Diameter. Determine the diameter of 10 strands of suture as directed under "Sutures-Diameter" in the USP. The average of the 30 measurements of the 10 strand sample shall be within the limits of the diameter range for the size being measured. In no case shall an individual measurement be less than the upper control point for the next smaller size, nor greater than the mid-point for the next largest size. Twenty-eight (28) of the thirty (30) individual diameter measurements shall be within the parameters described herein.

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4.4.3 Tensile strength. Test for tensile strength shall be performed on 10 strands as specified in the USP and shall comply with the requirements of columns (A) and (B) of Limits on Knot-Pull Tensile Strength specified in 3.2.3. If one of these ten strands is below the minimum strength specified in column (B), 20 new strands from the same lot of sutures shall be tested for tensile strength as specified above. Each of these 20 strands shall comply with the requirments of column (B) of Limits on Knot-pull Tensile Strength in 3.2.3. Therefore, of the 30 measurements not more than one shall be below the requirements of column (B).

4.4.4 Length. Test for length shall be conducted as specified in the USP.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One spool containing one hundred (100) yards as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit package. Each spool shall be wrapped in transparent plastic material or packaged in a two piece plastic container consisting of a solid plastic base and a transparent plastic cover through which the label of the spool shall be readily visible. Wrap or container shall be secured to prevent accidental opening.

5.1.1.1.2 Intermediate package. Twelve (12) units of issue shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, class domestic or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A. Two hundred eighty eight (288) units of issue shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

5.2.2 Level B. Two hundred eighty eight units of issue shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 2, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. No marking required. Label on spool shall be visible through wrap or container (see 3.4).

5.3.2 Intermediate packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. Lot (control) number shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown.

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6. NOTES

6.1 Intended use. The intended use of the items procured under the requirements of this specification is for general closure of open wounds during surgical procedures.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, NSN, number and date of this specification.
b. Applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following sizes of braided silk sutures wound on spools, as listed in the Federal Supply Catalog:

National Stock No. (NSN)	Item Identification
6515-00-721-9776	SUTURE, NONABSORBABLE, SURGICAL, Silk, Braided, Size 0, 100 yards.
6515-00-894-2968	SUTURE, NONABSORBABLE, SURGICAL, Silk, Braided, Size 2, 100 yards.
6515-00-894-2969	SUTURE, NONABSORBABLE, SURGICAL, Silk, Braided, Size 3, 100 yards.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:
Defense Supply Agency - DM

Project No. 6515-1906
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-37128 SUTURE, NONABSORBABLE, SURGICAL

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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